

☐ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number: 15052.002.0
Submittal Title: Wall Sleeves and Link Seals
TO: County of Hawaii, DEM ☐ For Information Only
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec: 15052	Paragraph: 1.04A, 1.04B, 2.02A, 2.02B, 3.01B
Authorized By: Krishna Dubbudu	Date Submitted: 09/26/2025

Submittal Certification	
Check Either (A) or (B):	
☒ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title: Krishna Dubbudu, Project Engineer	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/29/2025
SPECIFICATION SECTION # 15052
SPECIFICATION COMMON WORK RESULTS
PARAGRAPH 1.04-B
DRAWING N/A
SUBCONTRACTOR N/A
SUPPLIER C&B Piping
MANUFACTURER N/A

CERTIFIED BY CQCM or Designee: 

SECTION 15052

COMMON WORK RESULTS FOR GENERAL PIPING

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes: Basic materials and methods for metallic and plastic piping systems.

1.02 REFERENCES

- A. American Society of Mechanical Engineers (ASME):
 - 1. B16.5 - Pipe Flanges and Flanged Fittings: NPS 1/2 Through 24.
 - 2. B16.47 - Large Diameter Steel Flanges: NPS 26 Through NPS 60 Metric/Inch Standard.
- B. American Water Works Association (AWWA):
 - 1. C11 - Rubber-Gasket Joints for Ductile-Iron Pressure Pipe.
 - 2. C105 - Polyethylene Encasement for Ductile-Iron Pipe Systems.
 - 3. C111 - Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings.
 - 4. C151 - Ductile-Iron Pipe, Centrifugally Cast.
 - 5. C207 - Standard for Steel Pipe Flanges for Waterworks Services-Size 4 In. Through 144 In.
- C. ASTM International (ASTM):
 - 1. A74 - Standard Specification for Cast Iron Soil Pipe and Fittings.
 - 2. A193 - Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications.
 - 3. A194 - Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both.
 - 4. A307 - Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60,000 PSI Tensile Strength.
 - 5. A320 - Standard Specification for Alloy-Steel and Stainless Steel Bolting for Low-Temperature Service.
 - 6. A563 - Standard Specification for Carbon and Alloy Steel Nuts.
 - 7. B88 - Standard Specification for Seamless Copper Water Tube.
 - 8. F37 - Standard Test Methods for Sealability of Gasket Materials.
 - 9. F593 - Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.
 - 10. F594 - Standard Specification for Stainless Steel Nuts.
 - 11. F2329 - Standard Specification for Zinc Coating, Hot-Dip, Requirements of Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts, and Special Threaded Fasteners.

- D. California Health and Safety Code.
- E. NSF International (NSF):
 - 1. 61 – Drinking Water System Components – Health. Effects.
 - 2. 372 - Drinking Water System Components - Lead Content.

1.03 DEFINITIONS

- A. Buried pipes: Pipes that are buried in the soil with or without a concrete pipe encasement.
- B. Exposed pipe: Pipes that are located above ground, or located inside a structure, supported by a structure, or cast into a concrete structure.
- C. Underground pipes: Buried pipes - see A. above.
- D. Underwater pipes: Pipes below the top of walls in basins or tanks containing water.
- E. Wet wall: A wall with water on at least 1 side.
- F. Pipes adjacent to a wet wall: Pipe centerline within 10 inches of the wet wall.

1.04 SUBMITTALS

- ✓ A. Submit as specified in Section 01330 - Submittal Procedures.
- ✓ B. Product data:  Data provided for Wall Sleeves & Seals
 - 1. For each piping product in this Section as applicable:
 - a. Design features.
 - b. Load capacities.
 - c. Material designations by UNS alloy number or ASTM Specification and Grade.
 - d. Data needed to verify compliance with the Specifications.
 - e. Catalog data.
 - f. Clearly mark submittal information to show specific items, materials, and accessories or options being furnished.
 - g. Detailed piping layout drawings.  Any of these products are not in Contact with drinking water
- C. Calculations: 
 - 1. Provide calculations in accordance with NSF 372 for materials in contact with drinking water.

PART 2 PRODUCTS

2.01 GENERAL

- A. Pipes:
 - 1. Provide new pipe.

2. Piping provided from manufacturers/distributors inventory is subject to the following condition:
 - a. Provide proof pipe manufactured more than 6 months prior to delivery was stored properly and the material and/or coating was not subjected to ultraviolet (UV) degradation.
3. Mark each piece of pipe in accordance with applicable standards.

✓ 2.02 LINK TYPE SEALS

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- ✓ A. Characteristics:
 1. Modular mechanical type, consisting of interlocking neoprene or synthetic rubber links shaped to continuously fill the annular space between the pipe and wall opening.
 2. Links to form a continuous rubber belt around the pipe.
 3. Provide a nylon polymer pressure plate with Type 316 stainless steel hardware. Isolate pressure plate from contact with wall sleeve.
 4. Hardware to be Type 316 stainless steel.
 - a. Provide anti-galling lubricant for threads.
- ✓ B. One of the following or equal:
 1. ~~Link-Seal.~~
 2. ~~Pipe Linx.~~

2.03 FLANGE BOLTS AND NUTS

- A. General:
 1. Washer:
 - a. Provide a washer for each nut.
 - b. Washer shall be of the same material as the nut.
 2. Nuts: Heavy hex-head.
 - a. Stainless steel nuts shall be finished with TRIPAC 2000 coating system to minimize galling and provide proper torque.
 3. Cut and finish flange bolts to project:
 - a. Face of the bolt shall exceed face of nut by 1/8 inch minimum.
 - b. A maximum of 1/4 inch beyond outside face of nut after assembly.
 4. Tap holes for cap screws or stud bolts when used.
 5. Lubricant for stainless steel bolts and nuts:
 - a. Chloride-free.
 - b. Manufacturers: One of the following or equal:
 - 1) Huskey FG-1800 Anti-Seize.
 - 2) Weicon Anti-Seize High-Tech.
- B. Ductile iron pipe:
 1. On exposed pipes with pressures equal to or less than 150 pounds per square inch gauge (psig):
 - a. Bolts: In accordance with ASTM A307, Grade B.
 - b. Nuts: In accordance with ASTM A563, Grade A.
 - c. Bolts and Nuts: Hot-dip galvanized in accordance with ASTM F2329.
 2. On exposed pipes with pressures greater than 150 psig:
 - a. Bolts: In accordance with ASTM A193, Grade B7.

- b. Nuts: In accordance with ASTM A194, Grade 2H.
 - c. Bolts and nuts: Hot-dip galvanized in accordance with ASTM F2329.
 - 3. On underwater pipes and pipes adjacent to wet walls:
 - a. Bolts: In accordance with ASTM A193, Grade B8M.
 - b. Nuts: In accordance with ASTM A194, Grade 8M.
 - 4. On buried pipes:
 - a. Bolts: In accordance with ASTM A193, Grade B8M.
 - b. Nuts: In accordance with ASTM A194, Grade 8M for nuts.
 - c. Encase in 2 layers of loose polyethylene wrap in accordance with AWWA C105.
- C. Plastic pipe:
- 1. On exposed pipes:
 - a. Bolts: In accordance with ASTM A307, Grade B.
 - b. Nuts: In accordance with ASTM A563, Grade A.
 - c. Bolts and Nuts: Hot-dip galvanized in accordance with ASTM F2329.
 - 2. On underwater pipes and pipes adjacent to wet walls:
 - a. Bolts: In accordance with ASTM A193, Grade B8M.
 - b. Nuts: In accordance with ASTM A194, Grade 8M.
- D. Steel pipe:
- 1. On exposed pipes:
 - a. For ASME B16.5 Class 150 flanges and AWWA C207 Class D flanges:
 - 1) Bolts: In accordance with ASTM A307, Grade B.
 - 2) Nuts: In accordance with ASTM A563, Grade A.
 - 3) Bolts and Nuts: Hot-dip galvanized in accordance with ASTM F2329.
 - b. For ASME B16.5 and B16.47 Class 300 flanges and AWWA C207 Class E and F flanges:
 - 1) Bolts: In accordance with ASTM A193, Grade B7.
 - 2) Nuts: In accordance with ASTM A194, Grade 2H.
 - 2. On underwater pipes and pipes adjacent to wet walls:
 - a. Bolts: In accordance with ASTM A193, Grade B8M.
 - b. Nuts: In accordance with ASTM A194, Grade 8M.
 - 3. Low-temperature service: Stainless steel, Type 316:
 - a. Bolts: In accordance with ASTM A320, Grade B8M, Class 1, heavy hex.
 - b. Nuts: In accordance with ASTM A194, Grade B8M, heavy hex.
 - c. Washers: Alloy group matching bolts and nuts.
 - 4. High-temperature service or high-pressure service: Stainless steel, Type 316:
 - a. Bolts: In accordance with ASTM A193, Grade B8M, Class 1, heavy hex.
 - b. Nuts: In accordance with ASTM A194, Grade 8, heavy hex.
 - c. Washers: Alloy group matching bolts and nuts.

2.04 MECHANICAL JOINTS BOLTS AND NUTS

- A. Bolts including T-Bolts:
 - 1. Type 316 stainless steel in accordance with ASTM F593.
- B. Heavy Hex Nuts:
 - 1. Type 316 stainless steel in accordance with ASTM F594, finished with TRIPAC 2000 coating system to minimize galling and provide proper torque.

2.05 GASKETS

- A. General.
 - 1. Gaskets shall be suitable for the specific fluids, pressure, and temperature conditions.
 - 2. Capable of being applied on surface of piping with cavities to provide for an improved seal with the internal piping pressure.
- B. Gaskets for flanged joints in ductile iron and steel piping for sewage service:
 - 1. Suitable for pressures equal and less than 150 pounds per square inch gauge, temperatures equal and less than 250 degrees Fahrenheit, and raw sewage service.
 - 2. Gasket material:
 - a. SBR or neoprene elastomer with minimum Shore A hardness value of 70.
 - b. Reinforcement: Cloth or synthetic fiber.
 - c. Thickness: Minimum 3/32-inch thick for less than 10-inch pipe; minimum 1/8 inch thick for 10-inch and larger pipe.
 - 3. Manufacturers: One of the following or equal:
 - a. Pipe less than 48 inches in diameter:
 - 1) Garlock, Style 7797.
 - 2) John Crane, similar product.
 - b. Pipe 48 inches in diameter and larger:
 - 1) Garlock, Style 3760.
 - 2) John Crane, similar product.
- C. Gaskets for grooved end ductile iron and steel piping:
 - 1. Suitable for pressures equal to the encapsulating coupling or flange adapter.
 - 2. Material: Pressure responsive elastomer.
 - a. Ductile iron piping: FlushSeal® type.
 - 1) Halogenated Butyl: Grade M; for temperatures to 200 degrees Fahrenheit.
 - 2) Nitrile: Grade S; for temperatures to 180 degrees Fahrenheit.
 - b. Steel piping:
 - 1) EPDM: Grade E; for temperatures to 230 degrees Fahrenheit.
 - 2) EPDM-HP: Grade EHP, for temperatures to 250 degrees Fahrenheit.
 - 3) Nitrile: Grade T, for temperatures to 180 degrees Fahrenheit.
 - 3. Gaskets shall be verified as suitable for the intended service.
 - a. Temperature ratings may vary depending on the fluid/media.
 - 4. Manufacturers: Gaskets shall be of the same manufacturer as the encapsulating couplings/flange adapters.
 - a. Victaulic Company.
- D. Gaskets for flanged joints in polyvinyl chloride and polyethylene piping:
 - 1. Suitable for pressures equal to and less than 150 pounds per square inch gauge, with low flange bolt loadings, temperatures equal to and less than 120 degrees Fahrenheit, and polymer, chlorine, caustic solutions, and other chemicals, except chemicals which liberate free fluorine including fluorochemicals and gaseous fluorine.
 - 2. Material:
 - a. Chemical systems: 0.125-inch thick Viton™ rubber.
 - b. Sewer and water: 0.125-inch thick SBR.

3. Manufacturers: One of the following or equal:
 - a. Garlock.
 - b. John Crane, similar product.
- E. Gaskets for flanged joints in gas or liquefied petroleum gas piping:
 1. Digester gas in stainless steel, or black steel piping: Suitable for pressures equal to and less than 150 pounds per square inch gauge, temperatures equal to and less than 200 degrees Fahrenheit, and digester gas and mild acid concentrations.
 2. Chlorine gas application in black steel piping: Suitable for pressures equal to or less than 300 pounds per square inch gauge, temperatures equal to or less than 100 degrees Fahrenheit, and chlorine gas application.
 3. Liquefied petroleum, propane, and natural gas applications in black steel piping: Suitable for pressures equal to and less than 250 pounds per square inch gauge, temperatures equal to and less than 100 degrees Fahrenheit, and liquefied petroleum gas, propane gas, and natural gas application.
 4. Material:
 - a. Microcellular Teflon™ outer layers with rigid center layer.
 - b. Sealability in accordance with ASTM F37, less than 0.55 millimeters per hour leakage of iso-octane at 1,000 pounds per square inch gasket load and 9.8 pounds per square inch fluid pressure.
 5. Manufacturers: One of the following or equal:
 - a. Garlock, Style 3545.
 - b. John Crane, similar product.
- F. Gaskets for flanged joints in low pressure air piping:
 1. Suitable for pressures equal to and less than 150 pounds per square inch gauge, temperatures equal to and less than 300 degrees Fahrenheit, and compressed air service.
 2. Material: EPDM elastomer, 1/8-inch thick, 60 Shore hardness, smooth surface.
 3. Manufacturers: One of the following or equal:
 - a. Garlock, Style 8314.
 - b. John Crane, similar product.
- G. Gaskets for flanged joints in ductile iron or steel water piping:
 1. Suitable for hot or cold water, pressures equal to and less than 150 pounds per square inch gauge, and temperatures equal to and less than 160 degrees Fahrenheit.
 2. Material:
 - a. SBR or neoprene elastomer, compressed, with non-asbestos fiber reinforcement.
 3. Manufacturers: One of the following or equal:
 - a. Garlock, Bluegard 3300.
 - b. John Crane, similar product.
- H. Gaskets for flanged joints in ductile iron or steel drinking water piping meeting NSF 61 requirements:
 1. Suitable for hot or cold water, pressures equal to or less than 150 pounds per square inch gauge, and temperatures equal to or less than 160 degrees Fahrenheit.

2. Material:
 - a. EPDM material with 80 Shore A durometer rating.
3. Manufacturers: One of the following or equal:
 - a. Garlock, 98206.
 - b. John Crane, similar product.

PART 3 EXECUTION

3.01 INSTALLATION

- A. General:
 1. Piping drawings:
 - a. Except in details, piping is indicated diagrammatically. Not every offset and fitting, or structural difficulty that may be encountered has been indicated on the Drawings. Sizes and locations are indicated on the Drawings.
 - b. Perform minor modifications to piping alignment where necessary to avoid structural, mechanical, or other type of obstructions that cannot be removed or changed.
 - 1) Modifications are intended to be of minor scope, not involving a change to the design concept or a change to the Contract Price or Contract Times.
 2. Piping alternatives:
 - a. Provide piping as specified in this Section, unless indicated on the Drawings or specified otherwise.
 - b. Alternative pipe ratings:
 - 1) Piping with greater pressure rating than specified may be substituted in lieu of specified piping without changes to the Contract Price.
 - 2) Piping of different material may not be substituted in lieu of specified piping.
 - c. Valves in piping sections: Capable of withstanding specified test pressures for piping sections and fabricated with ends to fit piping.
 - d. Grooved joints: Use couplings, flange adapters, and fittings of the same manufacturer.
 - 1) Manufacturer's factory trained representative:
 - a) Provide on-site training for Contractor's field personnel.
 - b) Periodically visit the jobsite to verify Contractor is following best recommended practices.
 - 2) Distributor's representative is not considered qualified to conduct the training or jobsite visits.
 - e. Flanged joints: where 1 of the joining flanges is raised face type, provide a matching raised face type flange for the other joining flange.
 - 1) Wire brush and clean flange faces to remove all oil, grease, loose primer, mill scale, and any other foreign matter that could affect the proper seating of the gasket.
 3. Unless otherwise indicated on the Drawings, piping at pipe joints, fittings, couplings, and equipment shall be installed without rotation, angular deflection, vertical offset, or horizontal offset.



B.

Wall and slab penetrations:

1. Provide sleeves for piping penetrations through aboveground masonry and concrete walls, floors, ceilings, roofs, unless specified or otherwise indicated on the Drawings.
2. For piping 1 inch in nominal diameter and larger, provide sleeves with minimum inside diameters of 1 inch plus outside diameter of piping. For piping smaller than 1 inch in nominal diameter, provide sleeve of minimum twice the outside diameter of piping.
 - a. Arrange sleeves and adjacent joints so piping can be pulled out of sleeves and replaced without disturbing the structure.
 - b. Cut ends of sleeves flush with surfaces of concrete, masonry, or plaster.
 - c. Conceal ends of sleeves with escutcheons where piping runs through floors, walls, or ceilings of finished spaces within buildings.
 - d. Seal spaces between pipes and sleeves with link-type seals when not otherwise specified or indicated on the Drawings.
3. Provide flexibility in piping connecting to structures to accommodate movement due to soil settlement and earthquakes. Provide flexibility using details indicated on the Drawings.
4. Core drilled openings:
 - a. Do not damage or cut existing reinforcing bars, electrical conduits, or other items embedded in the existing concrete without acceptance by Engineer.
 - b. Determine location of reinforcing bars or other obstructions with a non-destructive indicator device.
 - c. Remove dust and debris from hole using compressed air.

C. Exposed piping:

1. Install exposed piping in straight runs parallel to the axes of structures, unless otherwise indicated on the Drawings:
 - a. Install piping runs plumb and level, unless otherwise indicated on the Drawings.
 - 1) Slope plumbing drain piping with a minimum of 1/4-inch per foot downward in the direction of flow.
2. Install exposed piping after installing equipment and after piping and fitting locations have been determined.
3. Support piping: As specified in Sections 15061 - Pipe Supports, 15062 - Preformed Channel Pipe Support System, and 15063 - Non-Metallic Pipe Support System:
 - a. Do not transfer pipe loads and strain to equipment.
4. In addition to the joints indicated on the Drawings, provide unions, flexible couplings, flanged joints, flanged coupling adapters, and other types of joints or means which are compatible with and suitable for the piping system, and necessary to allow ready assembly and disassembly of the piping.
5. Assemble piping without distortion or stresses caused by misalignment:
 - a. Match and properly orient flanges, unions, flexible couplings, and other connections.
 - b. Do not subject piping to bending or other undue stresses when fitting piping.
 - c. Do not correct defective orientation or alignment by distorting flanged joints or subjecting flange bolts to bending or other undue stresses.

- d. Flange bolts, union halves, flexible connectors, and other connection elements shall slip freely into place.
 - e. Alter piping assembly to fit when proper fit is not obtained.
 - f. Install eccentric reducers or increasers with the top horizontal for pump suction piping.
- D. Buried piping:
 - 1. Bury piping with minimum 3-foot cover without air traps, unless otherwise indicated on the Drawings.
 - 2. Where 2 similar services run parallel to each other, piping for such services may be laid in the same trench.
 - a. Lay piping with sufficient room for assembly and disassembly of joints, for thrust blocks, for other structures, and to meet separation requirements of public health authorities having jurisdiction.
 - 3. Laying piping:
 - a. Lay piping in finished trenches free from water or debris. Begin at the lowest point with bell ends up slope.
 - b. Place piping with top or bottom markings with markings in proper position.
 - c. Lay piping on an unyielding foundation with uniform bearing under the full length of barrels.
 - d. Where joints require external grouting, banding, or pointing, provide space under and immediately in front of the bell end of each section laid with sufficient shape and size for grouting, banding, or pointing of joints.
 - e. At the end of each day's construction, plug open ends of piping temporarily to prevent entrance of debris or animals.
 - 4. Concrete encase buried pipe installed under concrete slabs or structures.
- E. Venting piping under pressure:
 - 1. Lay piping under pressure flat or at a continuous slope without air traps, unless otherwise indicated on the Drawings.
 - 2. Install plug valves as air bleeder cocks at high points in piping.
 - a. Provide 1-inch plug valves for water lines, and 2-inch plug valves for sewage and sludge lines, unless otherwise indicated on the Drawings.
 - 3. Provide additional pipe taps with plug cocks and riser pipes along piping as required for venting during initial filling, disinfecting, and sampling.
 - 4. Before piping is placed into service, close plug valves and install plugs. Protect plugs and plug valves from corrosion in as specified in Section 09960 - High-Performance Coatings.
- F. Condensate in digester gas piping:
 - 1. Slope digester gas piping to drip traps or low-point drains at minimum 1/2-inch per foot where condensate flows against the gas or 1/4-inch per foot where condensate flows with gas.
 - 2. Install tapered filler pieces between flanges at high points of straight runs to provide for slope reversals.
 - a. Do not subject piping to high stresses in order to change direction.
 - 3. Provide pipe taps, threaded nipples, and 1-inch plug valves at low points in concrete utility boxes with lids.

- G. Restraining buried piping:
 - 1. Restrain piping at valves and at fittings where piping changes direction, changes sizes, and at ends:
 - a. When piping is underground, use mechanical restraints or push-on restraints.
 - 1) Concrete thrust blocks shall not be used unless specifically called out with a note on the Drawings or specified.
 - b. Determine thrust forces by multiplying the nominal cross-sectional area of the piping by design test pressure of the piping.
 - 2. Provide restraints with ample size to withstand thrust forces resulting from test pressures:
 - a. During testing, provide suitable temporary restraints where piping does not require permanent restraints.
 - 3. Where thrust blocks are allowed, place concrete thrust blocks against undisturbed soil.
 - 4. Where thrust blocks are allowed, place concrete so piping joints, fittings, and other appurtenances are accessible for assembly and disassembly.
 - 5. Provide underground mechanical restraints where specified in Attachment A - Piping Schedule.
- H. Restraining above ground and underwater piping:
 - 1. Restrain piping at valves and at fittings where piping changes direction, changes sizes, and at ends:
 - a. When piping is aboveground or underwater, use mechanical or structural restraints.
 - b. Determine thrust forces by multiplying the nominal cross-sectional area of the piping by design test pressure of the piping.
 - 2. Provide restraints with ample size to withstand thrust forces resulting from test pressures:
 - a. During testing, provide suitable temporary restraints where piping does not require permanent restraints.
- I. Connections to existing piping:
 - 1. Expose existing piping to which connections are to be made with sufficient time to permit, where necessary, field adjustments in line, grade, or fittings:
 - a. Protect domestic water/potable water supplies from contamination:
 - 1) Make connections between domestic water supply and other water systems in accordance with requirements of public health authorities.
 - 2) Provide devices approved by Owner of domestic water supply system to prevent flow from other sources into the domestic supply system.
 - 2. Make connections to existing piping and valves after sections of new piping to be connected have been tested and found satisfactory.
 - 3. Provide sleeves, flanges, nipples, couplings, adapters, and other fittings needed to install or attach new fittings to existing piping and to make connections to existing piping.
 - 4. For flanged connections, provide stainless steel bolts with isolation bushings and washers, and full-face flange gaskets.

- J. Connections between ferrous and nonferrous metals:
 - 1. Connect ferrous and nonferrous metal piping, tubing, and fittings with dielectric couplings especially designed for the prevention of chemical reactions between dissimilar metals.
 - 2. Nonferrous metals include aluminum, copper, and copper alloys.
- K. Flanged connections between dissimilar metals such as ductile iron pipe and steel pipe:
 - 1. Provide stainless steel bolts with isolation bushings and washers, and full-face flange gaskets.
- L. Where stainless steel bolts and/or nuts are in contact with non-stainless steel metals, use isolation bushings and washers.

3.02 CLEANING

- A. Piping cleaning:
 - 1. Upon completion of installation, clean piping interior of foreign matter and debris.
 - 2. Perform special cleaning when required by the Contract Documents.
- B. Cleaning potable water piping:
 - 1. Flush and disinfect potable water piping as specified in Section 01757 - Disinfection, and other regulatory requirements.
- C. Cleaning air piping:
 - 1. Perform special cleaning of filtered air piping from the intake clean air plenums to the discharge points and high-pressure air piping.
 - a. Protect surfaces from contamination.
 - 2. Special cleaning shall include wire brushing, power tool cleaning, wiping down with lint-free cloths, brooming, and vacuuming to remove rust, scale, weld spatter, dust, dirt, oil, and other matter deleterious to operation of the air system:
 - a. Do not sandblast installed piping.
 - 3. To the greatest extent possible, clean piping immediately prior to final closure of piping systems:
 - a. Enter piping, clean and wipe down surfaces, and vacuum out residue.
 - b. Clean surfaces not accessible to this cleaning operation after installation within 6 hours preceding installation.
 - 4. Subsequent to cleaning, protect surfaces from contamination by dust, dirt, construction debris, and moisture, including atmospheric moisture:
 - a. Whether or not pipe upstream has been cleaned, temporarily seal openings in partially completed work except when installation is actively in progress.
 - b. When installation is actively in progress, seal openings at the end of each day's construction or when construction is temporarily stopped.
 - 5. Suspend cleaning and seal openings when inclement weather, including dust storms, is imminent.
 - 6. Use clean, dry air for testing the piping and other elements of the system.

7. Prior to introduction of air to the system, blow piping clean.
 - a. Blow with maximum discharge rate possible for minimum 4 hours, using new blowers or compressors and filters.
 8. Clean surfaces that become contaminated prior to acceptance.
- D. Cleaning steam piping:
1. Thoroughly clean steam piping, boiler feed water piping, and condensate return piping by flushing with water.
 - a. Conduct cleaning operation after piping connections are completed and after piping has successfully passed pressure test.
 2. Utilize potable quality water, softened to remove hardness, for flushing water.
 - a. Provide temporary facilities and chemicals to add corrosion control additives to flushing water.
 - b. Utilize recommendations of boiler manufacturer for type and quantity of corrosion control additives.
 - c. When corrosion control additives are to be disposed of within treatment plant, use additives that are non-toxic to the treatment process.
 3. When the boiler system includes a feed water softening system, this system may be utilized as source of flushing water.
 - a. Provide adequately sized temporary storage tank when capacity of feed water softening system is less than required flow rate for flushing.
 4. Flush piping with water in sufficient quantities to produce a flow velocity of 6 feet per second in the main pipe and branch lines.
 5. Continue flushing until at least twice the volume of pipe has been flushed.
 6. Disconnect steam traps and piping less than 1/2-inch in nominal diameter prior to flushing operation.
 - a. After flushing operation is completed, drain entire system, and reconnect piping and steam traps.
 7. When cleaning operation is complete, remove temporary piping, pumps, tanks, and chemicals used to accomplish pipe cleaning.
 8. Prior to beginning cleaning operation, submit plan for the cleaning operation. Include details of equipment, procedures, and MSDS sheets for chemicals to be utilized.
- E. Conduct pressure and leak test, as specified.

3.03 COMMISSIONING

- A. As specified in Section 01756 - Commissioning and this Section.
- B. Field testing:
1. Piping system:
 - a. Witnessed.
 - b. Conduct pressure and leak test, as specified.

3.04 PIPING SCHEDULE

- A. Provide piping As specified in Attachment A - Piping Schedule unless noted otherwise in the Specifications or indicated on the Drawings with a written callout or note.

END OF SECTION

ATTACHMENT A - PIPING SCHEDULE

PIPING SCHEDULE

Process Abbrev.	Service	Nominal Diameter (inches)	Material	Class or Schedule	Pipe Spec. Section	Joints/ Fittings	Test Pressure/ Method	Lining	Coating	Comments
1W	Potable Water, County Supply									
	Buried	Under 4	PVC	SCH 80	15230	SW/PVC	200 psig/HH	None	None	Disinfect. WT for buried service.
		4 and over	DI	CL 51	15211	RMJ or RPO/DI	200 psig/HH	CM	AC and Double PEE	
	Exposed	Under 4	CU	See Spec	15281	Soldered/CU	150 psig/HH	None	EPP	
		4 and over	DI	CL 53	15211	FL or GE/DI	200 psig/HH	CM	EPP	
2W	Non-Potable Water									
	Buried	Under 4	PVC	SCH 80	15230	SW/PVC	200 psig/HH	None	None	
		4 and over	DI	CL 51	15211	RMJ or RPO/DI	200 psig/HH	CM	AC and Double PEE	
	Exposed	Under 4	GSP	SCH 40	15270	SC/300# MI	200 psig/HH	None	EPP	
		4 and over	DI	CL 53	15211	FL or GE/DI	200 psig/HH	CM	EPP	
	Underwater	All	PVC	SCH 80	15230	SW or FL/PVC	200 psig/HH	None	EPP	
3W	Chlorinated Final Effluent									
	Buried	Under 4	PVC	SCH 80	15230	SW/PVC	200 psig/HH	None	None	
		4 and over	DI	CL 51	15211	RMJ or RPO/DI	200 psig/HH	CM	AC and Double PEE	
	Exposed	Under 4	GSP	SCH 40	15270	SC/300# MI	200 psig/HH	None	EPP	
		4 and over	DI	CL 53	15211	FL or GE/DI	200 psig/HH	CM	EPP	
	Underwater	All	PVC	SCH 80	15230	SW or FL/PVC	200 psig/HH	None	EPP	
BP	Building Plumbing									
	Buried	All	CISP	ASTM A74	15254	Hub & Spigot/CISP	25 feet/GR	Coal Tar Pitch	EPP	For plumbing.
	Exposed	All	CISP	ASTM A74	15254	Hub & Spigot/CISP	25 feet/GR	Coal Tar Pitch	EPP	For toilets, sinks, and other plumbing.
BSL	Blended Sludge									
	Buried	All	DI	CL 51	15211	RMJ or RPO/DI	100 psig/HH	GL	AC and Double PEE	For exposed service, at connections to pumps and valves, use flanged ends only. All 90 degree bends shall be long-radius unless specifically noted otherwise on the Drawings.
	Exposed	All	DI	CL 53	15211	FL or GE/DI	100 psig/HH	GL	EPP	
CD	Chemical Drain									
	Buried	Under 4	PVC	SCH 80	15230	SW/PVC	25 psig/HH	None	None	
		4 and over	PVC AWWA C900	CL 150 (DR 18)	15230	PO/PVC AWWA C900 DR18	25 psig/HH	None	None	
	Exposed	All	CPVC	SCH 80	15230	SW or FL/CPVC	25 psig/HH	None	EPP	
CEN	Centrate									
	Buried	All	HDPE	SDR 21	15241	Welded	15 feet/GR	None	None	Remove inner weld
	Exposed	3 and less	PVC	SCH 80	15230	SW/PVC or FL	15 feet/GR	None	EPP	
		4 and over	HDPE	SDR 21	15241	Welded; FL at fittings	15 feet/GR	None	None	Remove inner weld; insulate exterior exposed
CLS	Chlorine Solution									
	Exposed	All	CPVC	SCH 80	15230, 15293	SW/CPVC	150 psig/HH	None	EPP	Double contained piping required where indicated on Drawings.
CSL	Cold Sludge									
	Buried	4 and over	DI	CL 51	15211	RMJ or RPO/DI	100 psig/HH	GL	AC and Double PEE	For exposed service, at connections to pumps and valves, use flanged ends only.
	Exposed	4 and over	DI	CL 53	15211	FL or GE/DI	100 psig/HH	GL	EPP	
	Underwater	4 and over	DI	CL 53	15211	FL/DI	100 psig/HH	GL	EPX-M-3	

Process Abbrev.	Service	Nominal Diameter (inches)	Material	Class or Schedule	Pipe Spec. Section	Joints/ Fittings	Test Pressure/ Method	Lining	Coating	Comments
D	Drain									
	Buried	Under 4	PVC	SCH 80	15230	SW/PVC	50 psig/HH	None	None	
		4 and over	DI	CL 51	15211	RMJ or RPO/DI	50 psig/HH	CM	AC and Double PEE	
	Exposed	Under 4	GSP	SCH 40	15270	SC/300# MI	50 psig/HH	None	EPP	
		4 and over	DI	CL 53	15211	FL or GE/DI	50 psig/HH	CM	EPP	
DG	Digester Gas									
	Buried	All	HDPE	Per Section 15241	15241	Socked Fusion per ASTM D2683 Butt Fusion per ASTM D3261	15 psig/LH	None	None	
	Exposed	All	SST	SCH 40S	15286	GE, or FL; FL where shown/SST	15 psig/LH	None	None	SST Grade per Section 15286
DSL	Digested Sludge									
	Buried	4 and Over	DI	CL51	15211	RMJ or RPO/DI	100 psig/HH	GL	AC and Double PEE	
	Exposed	4 and Over	DI	CL53	15211	FL or GE/DI	100 psig/HH	GL	EPP	For exposed service, at connections to pumps and valves, use flanged ends only.
	Underwater	4 and Over	DI	CL53	15211	FL/DI	100 psig/HH	GL	EPX-M-3	
FD	Floor Drain									
	Buried	Under 4	PVC	SCH 80	15230	SW/PVC	25 psig/HH	None	None	
		4 and Over	PVC AWWA C900	CL 150 (DR 18)	15230	PO/PVC AWWA C900 DR18	25 psig/HH	None	None	
	Exposed	All	DI	CL 53	15211	GE or FL/DI	25 feet/LH	CM	EPP	
FIL	Filtrate									
	Buried	4 and Over	DI	CL51	15211	RMJ or RPO/DI	25 feet/LH	CM	AC and Double PEE	
	Exposed	4 and Over	DI	CL 53	15211	FL or GE/DI	25 feet/LH	CM	EPP	For exposed service, at connections to pumps and valves, use flanged ends only.
FM	Force Main									
	Buried	Under 4	PVC	SCH 80	15230	SW/PVC	150 psig/HH	None	None	
		4 and Over	DI	CL 51	15211	RPO or RMJ/DI	150 psig/HH	CM	AC and Double PEE	
	Exposed	Under 4	CPVC	SCH 80	15230	SW, FL, or SC/CPVC	150 psig/HH	None	EPP	
		4 and Over	DI	CL 53	15211	GE or FL/DI	150 psig/HH	CM	EPP	
		18 and 24	316 SST	SCH 40S	15286	FL	150 psig/HH	None	None	Inside Screen Influent Channel and upstream of the channel as indicated on the Drawings.
FW	Fire Water									
	Buried	All	DI	CL 51	15211	RMJ or RPO/DI	175 psig/HH	CM	AC and Double PEE	Disinfect.
	Exposed	Under 4	GSP	SCH 40	15270	SC/300# MI	175 psig/HH	None	EPP	WT for buried service.
		4 and Over	DI	CL 53	15211	GE or FL/DI	175 psig/HH	CM	EPP	
GRR	Glycol Reservoir Return									
	Buried	Under 4	CU ASTM B88	Type K	15281	Soldered or Press Fit/CU ASME B16.51 ASME B16.22	150 psig/HH	None	None	EPDM gaskets: Insulate per Section 15082. WT for buried service
	Exposed	Under 4	CU ASTM B88	Type K	15281	Soldered or Press Fit/CU ASME B16.51 ASME B16.22	150 psig/HH	None	EPP	
GRS	Glycol Reservoir Supply									
	Buried	Under 4	CU ASTM B88,	Type K	15281	Soldered or Press Fit/CU ASME B16.51 ASME B16.22	150 psig/HH	None	None	EPDM gaskets: Insulate per Section 15082. WT for buried service
	Exposed	Under 4	CU ASTM B88,	Type K	15281	Soldered or Press Fit/CU ASME B16.51 ASME B16.22	150 psig/HH	None	EPP	

Process Abbrev.	Service	Nominal Diameter (inches)	Material	Class or Schedule	Pipe Spec. Section	Joints/ Fittings	Test Pressure/ Method	Lining	Coating	Comments
GRT	Grit Slurry									
	Buried	All	DI	CL 53	15211	RMJ or RPO/DI	50 psig/HH	GL	AC and Double PEE	For exposed service, use flanged ends at connections to pumps, valves, and where shown on Drawings only. All 90-degree bends shall be long-radius unless specifically noted otherwise on the Drawings.
	Exposed	All	DI	CL 53	15211	GE/DI	50 psig/HH	GL	EPP	
	Underwater	All	DI	CL 53	15211	FL/DI	50 psig/HH	GL	EPX-M-3	All 90 degree bends shall be long-radius unless specifically noted otherwise on the Drawings.
HFR, HFS	Hydraulic Fluid Return, Hydraulic Fluid Supply									
	Exposed	All	316 SST	0.120-inch thick (min)	15286	Per Specifications	4,000 psig	None	None	
HPA	High Pressure Air									
	Buried	All	BSP	SCH 40	15270	SC/300# MI	175 psig/AM	None	PCW	WT for buried service.
	Exposed	2 and under	BSP	SCH 40	15270	SC or W/300# MI	175 psig/AM	None	EPP	
		2 1/2 and over	BSP	SCH 40	15270	FL or GE/300# STL	175 psig/AM	None	EPP	
	Underwater	All	316L SST	SCH 40S	15286	SC or FL/316L SST	175 psig/AM	None	None	
HPW	High Pressure Water									
	Buried	All	304L SST	SCH 80S	15286	SCW/304L SST	3,000 psig/HH	None	I	Field welding is acceptable for this service provided it is performed by certified stainless steel welders. Insulate per Section 15082.
	Exposed	All	304L SST	SCH 80S	15286	SCW/304L SST	3,000 psig/HH	None	I	
HSL	Heated Sludge									
	Buried	4 and over	DI	CL 51	15211	RMJ or RPO/DI	100 psig/HH	GL	AC and Double PEE	For exposed service, at connections to pumps and valves, use flanged ends only.
	Exposed	4 and over	DI	CL 53	15211	FL or GE/DI	100 psig/HH	GL	EPP	
HW, HWR, HWS	Hot Water, Hot Water Return, Hot Water Supply									
	Buried	All	SST (Grade per Section 15286)	SCH 40S	15286	SC /SST	90 psig/HH	None	None	Insulate per Section 15082.
	Exposed	All	SST (Grade per Section 15286)	SCH 40S	15286	SC or FL /SST	90 psig/HH	None	None	
IA	Instrument Air									
	Buried	All	316 SST	0.035-inch thick (min)	15286	Swage/316 SST	175 psig/AM	None	None	WT for buried service.
	Exposed	All	316 SST	0.035-inch thick (min)	15286	Swage/316 SST	175 psig/AM	None	None	
	Underwater	All	316 SST	0.035-inch thick (min)	15286	Swage/316 SST	175 psig/AM	None	None	
LPG	Propane									
	Exposed	All	SST	SCH 40	15286	FL / SST	15 psig	None	None	SST Grade per Section 15286
MSL	Mixing Sludge									
	Buried	4 and over	DI	CL 51	15211	RMJ or RPO/DI	100 psig/HH	GL	AC and Double PEE	For exposed service, at connections to pumps and valves, use flanged ends only.
	Exposed	4 and over	DI	CL 53	15211	FL or GE/DI	100 psig/HH	GL	EPP	
	Underwater	4 and over	DI	CL 53	15211	FL/DI	100 psig/HH	GL	EPX-M-3	
NS	Nutrient Solution									
	Exposed	All	CPVC	SCH 80	15230	SW or FL/CPVC	150 psig/HH	None	None	

Process Abbrev.	Service	Nominal Diameter (inches)	Material	Class or Schedule	Pipe Spec. Section	Joints/ Fittings	Test Pressure/ Method	Lining	Coating	Comments
OF	Overflow									
	Buried	4 and Over	DIP	CL51	15211	RMJ or RPO/DI	15 feet/GR	GL	AC and Double PEE	Pothole and confirm material of existing 36-inch OF prior to fabricating pipe. If the existing 36-inch OF is not DIP, inform Engineer immediately and ask for direction on pipe material to use.
	Exposed	4 and Over	DIP	CL 53	15211	GE or FL/DI	15 feet/GR	GL	EPP	For exposed service, at connections to pumps and valves, use flanged ends only.
PD	Process Drain									
	Buried	Under 4	PVC	SCH 80	15230	SW/PVC	50 psig/HH	None	None	
		4 and Over	DI	CL 51	15211	RMJ or RPO/DI	50 psig/HH	CM	AC and Double PEE	Where not specified to be encased in concrete.
		4 and Over	PVC AWWA C900	CL 150 (DR 18)	15230	PO/PVC AWWA C900 DR18	25 psig/HH	None	None	Where encased in concrete under slabs.
	Exposed	Under 4	GSP	SCH 40	15270	SC/300# MI	50 psig/HH	None	EPP	
		4 and Over	DI	CL 53	15211	FL or GE/DI	50 psig/HH	CM	EPP	Use ceramic epoxy lining for process drains from grit washers and for vactor truck pipe.
PE	Primary Effluent									
	Buried	All	STL	Per 15272 Min 0.25"	15272	WLD	15 ft/GR	CM	CM and PTW	
PI	Primary Influent									
	Buried	All	DI	CI 51	15211	RMJ or RPO/DI	25 psig/HH	CM	AC and Double PEE	
	Exposed	All	DI	CL 53	15211	FL or GE/DI	25 psig/HH	CM	EPP	
POLY	Neat Polymer									
	Buried	All	PVC	SCH 80	15230	SW/PVC	50 psig/HH	None	None	
	Exposed	All	CPVC	SCH 80	15230	SW or Flanged/CPVC	50 psig/HH	None	WAE	
POS	Polymer Solution									
	Buried	All	PVC	SCH 80	15230	SW/PVC	175 psig/HH	None	None	
	Exposed	All	CPVC	SCH 80	15230	SW or Flanged/CPVC	175 psig/HH	None	WAE	
PR	Plant Recycle									
	Buried	Under 4	PVC	SCH 80	15230	SW/PVC	50 psig/HH	None	None	
		4 and Over	DI	CL 51	15211	RMJ or RPO/DI	50 psig/HH	CM	AC and Double PEE	
	Exposed	Under 4	GSP	SCH 40	15270	SC/300# MI	50 psig/HH	None	EPP	
		4 and Over	DI	CL 53	15211	FL or GE/DI	50 psig/HH	CM	EPP	
PS	Primary Sludge									
	Buried	All	DI	CL 51	15211	RMJ or RPO/DI	75 psig/HH	GL	AC and Double PEE	
	Exposed	All	DI	CL 53	15211	FL or GE/DI	75 psig/HH	GL	EPP	For exposed service, use flanged ends at connections to pumps, valves, and where shown on Drawings. All 90-degree bends shall be long-radius unless specifically noted otherwise on the Drawings.

Process Abbrev.	Service	Nominal Diameter (inches)	Material	Class or Schedule	Pipe Spec. Section	Joints/ Fittings	Test Pressure/ Method	Lining	Coating	Comments
PSC	Primary Scum									
	Buried	All	DI	CL 51	15211	RMJ or RPO/DI	75 psig/HH	GL	AC and Double PEE	
	Exposed	All	DI	CL 53	15211	FL or GE/DI	75 psig/HH	GL	EPP	For exposed service, use flanged ends at connections to pumps, valves, and where shown on Drawings. All 90-degree bends shall be long-radius unless specifically noted otherwise on the Drawings.
RD	Roof Drain									
	Exposed	Under 4	GSP	SCH 40	15270	SC/300# MI	25 feet/LH	None	EPP	
		4 and over	DI	CL 53	15211	FL or GE/DI	25 feet/LH	CM	EPP	
REC	Recirculation									
	Exposed	All	CPVC	SCH 80	15230	SW or FL/CPVC	75 psig/HH	None	EPP	
RSS	Return Secondary Sludge									
	Exposed	All	DI	CL 53	15211	FL	75 psig/HH	GL	EPP	
SAM	Sample									
	Buried	All	316 SST	SCH 40S	15286	FL/316SST	50 psig/HH	None	None	
	Exposed	All	316 SST	SCH 40S	15286	FL/316 SST	50 psig/HH	None	None	
SEP	Septage									
	Buried	All	PVC AWWA C900	CL 150 (DR 18)	15230	PO/PVC AWWA C900 DR18	25 psig/HH	None	None	Gravity
		All	DI	CL 53	15211	GE or FL/DI	100 psig/HH	GL	EPP	Pumped
	Exposed	Under 4	316 SST	SCH 40S	15286	SC or FL/316 SST	100 psig/HH	None	None	Inside Screen Influent Channel and upstream of the channels as indicated on the Drawings. All 90 degree bends shall be long-radius unless specifically noted otherwise on the Drawings.
		4 and Over	DI	CL 53	15211	GE or FL/DI	100 psig/HH	GL	EPP	
		4 and Over	316 SST	SCH 40S	15286	FL /316 SST	100 psig/HH	None	None	
	Underwater	Under 4	316 SST	SCH 40S	15286	SC or FL/316 SST	100 psig/HH	None	None	All 90 degree bends shall be long-radius unless specifically noted otherwise on the Drawings.
SPD	Sump Pump Discharge									
	Buried	Under 4	PVC	SCH 80	15230	SW/PVC	100 psig/HH	None	None	
		4 and Over	DI	CL 51	15211	RPO or RMJ/DI	100 psig/HH	CM	AC and Double PEE	
	Exposed	Under 4	316 SST	SCH 40S	15286	SC or FL /316 SST	100 psig/HH	None	None	
		4 and Over	DI	CL 53	15211	GE or FL/DI	100 psig/HH	CM	EPP	
		4 and Over	316 SST	SCH 40S	15286	FL /316 SST	100 psig/HH	None	None	Inside Screen Influent Channel, Screen Effluent Channel, and upstream of the channels as indicated on the Drawings.
	Underwater	Under 4	316 SST	SCH 40S	15286	SC or FL/316 SST	100 psig/HH	None	None	
SSC	Secondary Scum									
	Buried	All	DI	CL51	15211	RMJ or RPO/DI	75 psig/HH	GL	AC and Double PEE	
	Exposed	All	DI	CL 53	15211	FL	75 psig/HH	GL	EPP	For exposed service, use flanged ends at connections to pumps, valves, and where shown on Drawings. All 90-degree bends shall be long-radius unless specifically noted otherwise on the Drawings.

Process Abbrev.	Service	Nominal Diameter (inches)	Material	Class or Schedule	Pipe Spec. Section	Joints/ Fittings	Test Pressure/ Method	Lining	Coating	Comments
SW	Seal Water									
	Buried	All	PVC	SCH 80	15230	SW/PVC	200 psig/HH	None	None	
	Exposed	All	CU	See Spec	15281	Soldered/CU	150 psig/HH	None	EPP	
TD	Tank Drain									
	Exposed	All	DI	CL 53	15211	FL or GE/DI	100 psig/HH	CM	EPP	
TSL	Thickened Sludge									
	Buried	All	DI	CL51	15211	RMJ or RPO/DI	100 psig/HH	GL	AC and Double PEE	
	Exposed	All	DI	CL 53	15211	FL or GE/DI	100 psig/HH	GL	EPP	For exposed service, use flanged ends at connections to pumps, valves, and where shown on Drawings. All 90-degree bends shall be long-radius unless specifically noted otherwise on the Drawings.
V	Vent									
	Buried	All	CISP	ASTM A888		No Hub/CISP	25 ft/GR	Coal Tar Pitch	EPP	For plumbing.
	Exposed	All	CISP	ASTM A888		No Hub/CISP	25 ft/GR	Coal Tar Pitch	EPP	For toilets, sinks, and other plumbing.
WSS	Waste Secondary Solids									
	Buried	All	DI	CL 51	15211	RMJ or RPO/DI	75 psig/HH	CM	AC and Double PEE	
	Exposed	All	DI	CL 53	15211	FL	75 psig/HH	CM	EPP	
Notes:										
1. The following abbreviations used in the column of test method refer to the respective methods as specified in Section 15956.										
AM: Air method				HH: High head method				SC: Special case		
GR: Gravity method				LH: Low head method						
2. Abbreviations to designate piping include the following:										
AC: Asphaltic coating					FW: Fusion welded			psig: pounds per square inch gauge		
B&SP, BS: Bell and spigot					GA: Gauge, preceded by the designation			PTW: Polyethylene tape wrap		
BSP: Black steel pipe					GAL: Galvanized			PVC: Polyvinyl chloride		
CI: Cast iron					GE: Grooved end joint			RCCP: Reinforced concrete cylinder pipe, AWWA C300		
CISP: Cast iron soil pipe					GL: Glass lined			RMJ: Restrained Mechanical Joint		
CL: Class, followed by the designation					GSP: Galvanized steel pipe			RPO: Restrained Push-on		
CM: Cement mortar					I: Insulated			SC: Screwed-on		
CPVC: chlorinated polyvinyl chloride					JT: Joint test			SCH: Schedule, followed by the designation		
CTP: Coal tar pitch					MI: Malleable iron			SCW: Socket welded		
CU: Copper					MJ: Mechanical joint			SST: Stainless steel		
DG: double gasketed					NPS: Nominal pipe size, followed by the number in inches			STL: Steel		
DI: Ductile iron					PCW: Plastic coating or wrapping			SW: Solvent welded		
EP: Elastomeric polyurethane					PE: Polyethylene			VCP: Vitrified clay piping		
EPP: Epoxy polyurethane coating EPU-M-1, refer to Section 09960 - High-Performance Coatings.					PEE: Polyethylene encasement			W, WLD: Weld		
EPX-M-3: Epoxy, refer to Section 09960 - High-Performance Coatings.					PO: Push-on			WAE: Waterborne acrylic emulsion		
FL: Flange					psi: pounds per square inch			WT: Warning tape		

PRODUCT DATA



C&B PIPING (W), INC.

P.O. BOX 974

LEEDS, AL 35094

PHONE: (205) 699-0455

FAX: (205) 699-0773

www.cbpipling.com

DATE: 9/25/25

CONTRACTOR: NAN, INC.

PROJECT: HILO WWTP

BID DATE: 8/15/2024

WALL SLEEVE & LINK SEAL
SUBMITTAL

PLAN DETAILS P302 & P304

SPEC SECTION 15052-3

C&B PIPING WILL PROVIDE LAYOUT DRAWING SUBMITTALS SHOWING PIPE SYSTEMS AND ACTUAL PIPE SIZES, CLASSES, JOINTS PROPOSED FOR SUBMITTAL APPROVAL. THE DRAWINGS WILL BE SUBMITTED BY SYSTEM AS THEY ARE COMPLETED. THE SUBMITTAL PRODUCT LISTING AND ATTACHED CUT SHEETS ARE FOR REFERENCE TO COVER THE ITEMS THAT WILL BE SHOWN ON THE PIPING LAYOUT DRAWINGS.

WALL SLEEVES

PIPE PENETRATION WALL SLEEVES

1) PLAIN END CARBON STEEL WALL SLEEVES FOR CONCRETE PIPE PENETRATIONS

NOTE: PER PLAN DETAIL P302 - NO WALL COLLAR

2) PLAIN END 316LSS WALL SLEEVES FOR PIPE PENETRATIONS

NOTE: PER PLAN DETAIL P304 - SUPPLIED WITH WALL COLLAR

CUSTOM PIPING SYSTEMS

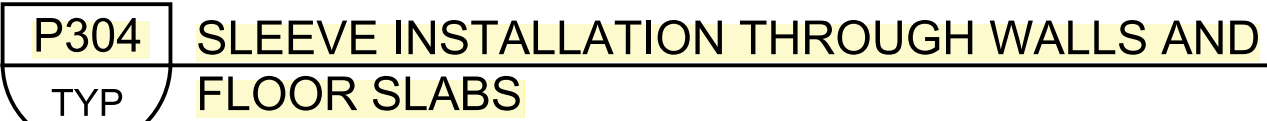
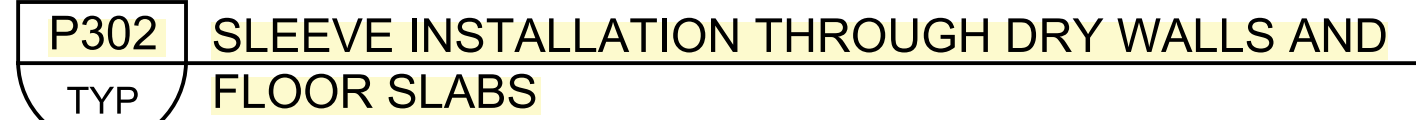
LINK TYPE SEALS

MECHANICAL SEAL SETS FOR WALL SLEEVE PIPE PENETRATIONS

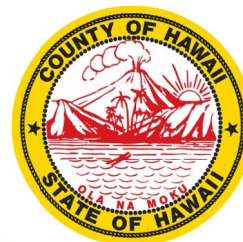
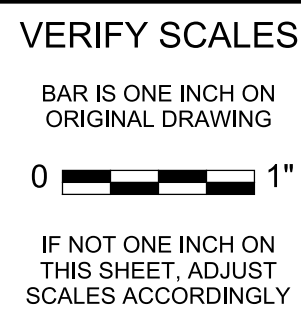
1) MECHANICAL SEAL SETS FOR WALL SLEEVE PIPE PENETRATIONS

NOTE: SUPPLIED W/316SS HARDWARE AND EPDM RUBBER

CCI PIPING SYSTEMS



DESIGNED
XHK
DRAWN
OB
CHECKED
WK
DATE
APR 2024



JOB NO. 201400
DRAWING NO. 00-T-05-706
SHEET NO. 961 of 1000

PLEASE SEND A FORM FOR EACH SIZE: PER PLANS DETAIL P302

Diameter:Contractor to Field Verify

Material:CARBON STEEL

Wall Thickness:SCH40 FOR SIZES UP

TO 6" AND 1/4" THICK FOR LARGER

SIZES.

Finishing: HOT DIPPED GALVANIZED_

Quantity:FIELD VERIFY *AIS DOMESTIC



WALL SLEEVES WITHOUT COLLAR

GENERAL NOTES

1. ALL WALL SLEEVES SHALL BE FABRICATED FROM STD. WT. PIPE A53 GRADE B OR A36 PLATE WITH THICKNESS MATCHING THAT OF STD. WT. PIPE AND SHALL BE HOT DIPPED GALVANIZED AFTER FABRICATION PER ASTM A123.
2. STAINLESS STEEL WALL SLEEVES SHALL BE FABRICATED FROM SCH 10S PIPE OR STAINLESS STEEL PLATE WITH THICKNESS MATCHING THAT OF SCH 10S PIPE AND SHALL BE PICKLED AND PASSIVATED AFTER FABRICATION PER ASTM A380.
3. ALL WALL COLLARS ON PIPES 12" OR SMALLER SHALL BE 1/4" THICK AND 2" WIDE
4. ALL WALL COLLARS ON PIPES 14" OR LARGER SHALL BE 1/4" THICK AND 3" WIDE
5. WALL PIPE LENGTH AND COLLAR PLACEMENT TO BE DETERMINED BY CUSTOMER
6. LENGTH AND COLLAR PLACEMENT TO BE DETERMINED BY CUSTOMER.
7. ALSO AVAILABLE IN PLAIN FINISH, COLD SPRAY GALVANIZED AND FUSION BOND EPOXY.

0	SUBMITTAL	06/18
No.	Revision / Issue	Date

Drawing Checked/Approved by:

STANDARD
WALL SLEEVES

DETAIL DRAWING

1.0



DRAWN BY: YBR

06/17/25

SCALE: N.T.S.

REV - 0

PLEASE SEND A FORM FOR EACH SIZE: PER PLANS DETAIL P304

Diameter:CONTRACTOR_TO_VERIFY

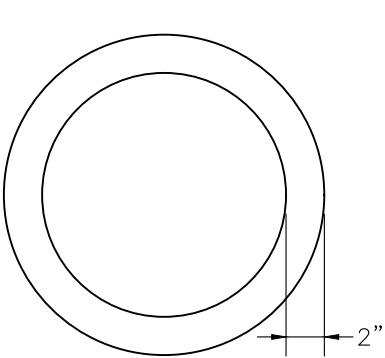
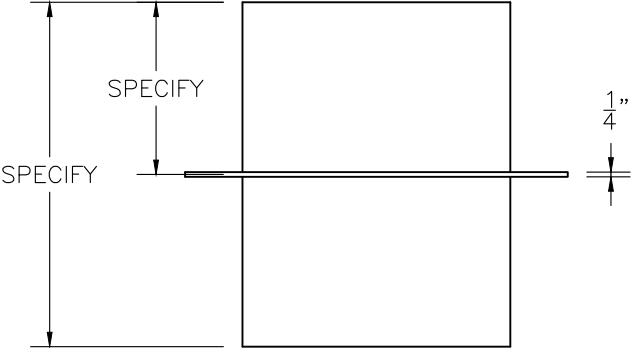
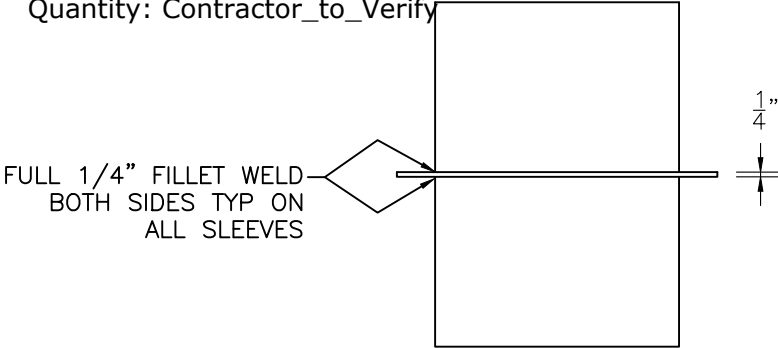
Material:316SS

Wall_Thickness:SCH40_6"&_SMALLER_

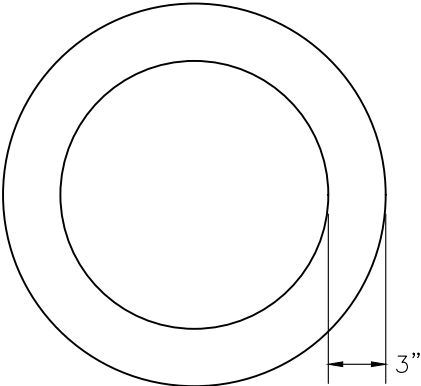
AND_1/4"_Thick_larger_then_6"

Finishing:AIS_Domestic_316SS

Quantity: Contractor_to_Verify



12" AND SMALLER WALL SLEEVES



14" AND LARGER WALL SLEEVES

GENERAL NOTES

1. CARBON STEEL WALL SLEEVES SHALL BE FABRICATED FROM STD. WT. PIPE A53 GRADE B OR A36 PLATE WITH THICKNESS MATCHING THAT OF STD. WT. PIPE, UNLESS OTHERWISE NOTED AND SHALL BE HOT DIPPED GALVANIZED AFTER FABRICATION PER ASTM A123.
2. STAINLESS STEEL WALL SLEEVES SHALL BE FABRICATED FROM SCH 10S PIPE OR STAINLESS STEEL PLATE WITH THICKNESS MATCHING THAT OF SCH 10S PIPE, UNLESS OTHERWISE NOTED AND SHALL BE PICKLED AND PASSIVATED AFTER FABRICATION PER ASTM A380.
3. ALL WALL COLLARS ON PIPES 12" OR SMALLER SHALL BE 1/4" THICK AND 2" WIDE.
4. ALL WALL COLLARS ON PIPES 14" OR LARGER SHALL BE 1/4" THICK AND 3" WIDE.
5. WALL PIPE LENGTH AND COLLAR PLACEMENT TO BE DETERMINED BY CUSTOMER.
6. LENGTH AND COLLAR PLACEMENT TO BE DETERMINED BY CUSTOMER.
7. ALSO AVAILABLE IN PLAIN FINISH, COLD SPRAY GALVANIZED AND FUSION BOND EPOXY.

0	SUBMITTAL	06/18
No.	Revision / Issue	Date

Drawing Checked/Approved by:

STANDARD
WALL SLEEVES

DETAIL DRAWING

1.0



DRAWN BY: YBR 06/17/25 SCALE: N.T.S. REV - 0

**PROPOSAL AND SPECIFICATIONS
FOR
HILO WASTEWATER TREATMENT PLANT (WWTP)
REHABILITATION AND REPLACEMENT PROJECT PHASE 1**

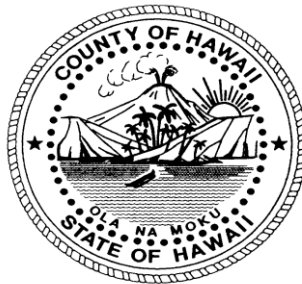
**WAIĀKEA, HILO, HAWAI‘I
COUNTY AND STATE OF HAWAI‘I
TMK: (3) 2-1-013: 002**

JOB NO. WW-4705R

JANUARY 2025
VOLUME 5 (DIVISION 00)

CONFORMED DOCUMENTS

These Conformed Documents unify addenda issued during the bid period. If discrepancies between the Conformed Documents and the Bid Documents are found, the Bid Documents with the original addenda shall govern.



County of Hawai‘i
Department of Environmental Management
Wastewater Division

345 Kekūanā‘o‘a, Suite 41
Hilo, Hawai‘i 96720

Prepared by:

Carollo Engineers, Inc.
733 Bishop Street, Suite 1610
Honolulu, HI, 96813

2. Piping provided from manufacturers/distributors inventory is subject to the following condition:
 - a. Provide proof pipe manufactured more than 6 months prior to delivery was stored properly and the material and/or coating was not subjected to ultraviolet (UV) degradation.
3. Mark each piece of pipe in accordance with applicable standards.

2.02 LINK TYPE SEALS

A. Characteristics:

1. Modular mechanical type, consisting of interlocking neoprene or synthetic rubber links shaped to continuously fill the annular space between the pipe and wall opening.
2. Links to form a continuous rubber belt around the pipe.
3. Provide a nylon polymer pressure plate with Type 316 stainless steel hardware. Isolate pressure plate from contact with wall sleeve.
4. Hardware to be Type 316 stainless steel.
 - a. Provide anti-galling lubricant for threads.

B. One of the following or equal:

1. ~~Link-Seal.~~
2. ~~Pipe Linx.~~

2.03 FLANGE BOLTS AND NUTS

A. General:

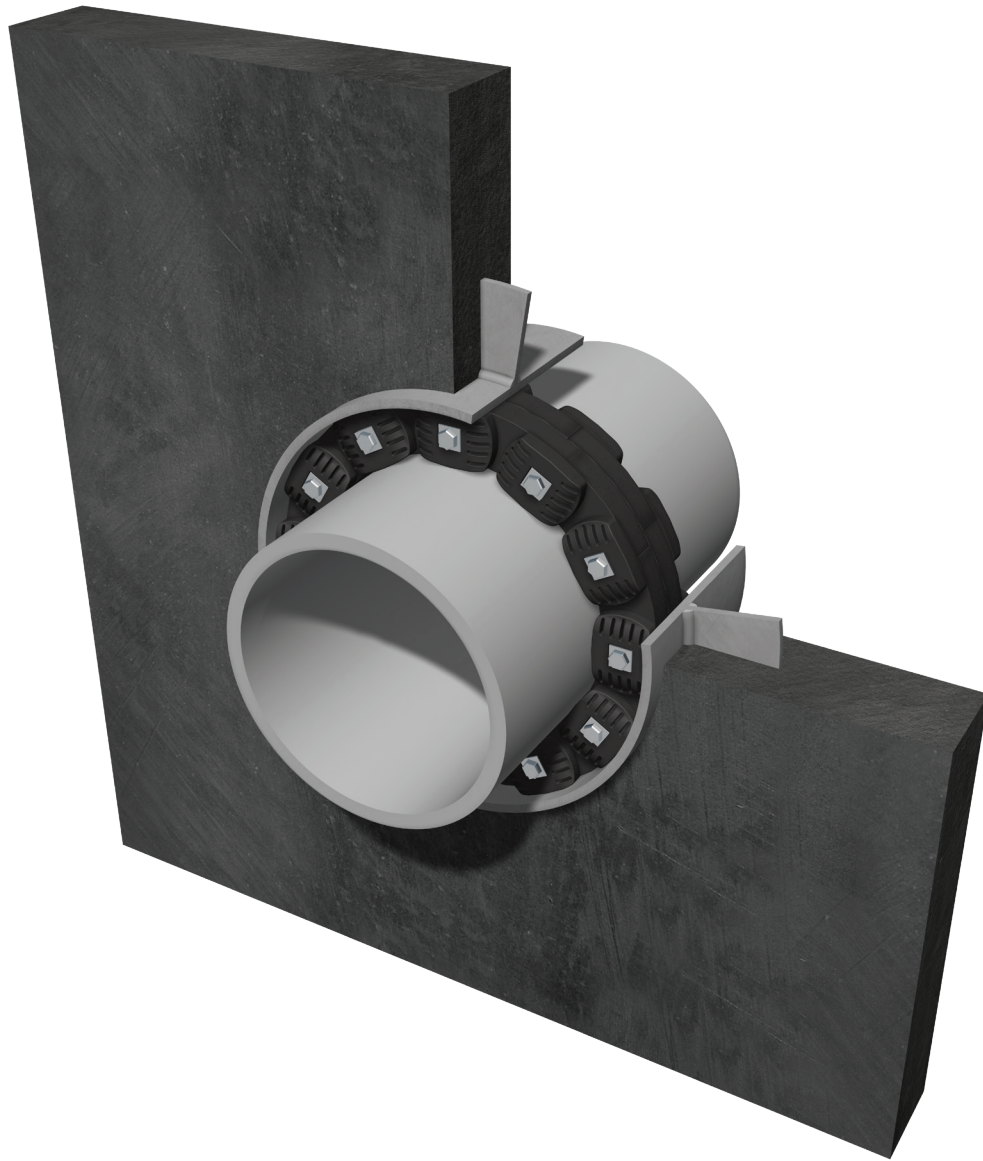
1. Washer:
 - a. Provide a washer for each nut.
 - b. Washer shall be of the same material as the nut.
2. Nuts: Heavy hex-head.
 - a. Stainless steel nuts shall be finished with TRIPAC 2000 coating system to minimize galling and provide proper torque.
3. Cut and finish flange bolts to project:
 - a. Face of the bolt shall exceed face of nut by 1/8 inch minimum.
 - b. A maximum of 1/4 inch beyond outside face of nut after assembly.
4. Tap holes for cap screws or stud bolts when used.
5. Lubricant for stainless steel bolts and nuts:
 - a. Chloride-free.
 - b. Manufacturers: One of the following or equal:
 - 1) Huskey FG-1800 Anti-Seize.
 - 2) Weicon Anti-Seize High-Tech.

B. Ductile iron pipe:

1. On exposed pipes with pressures equal to or less than 150 pounds per square inch gauge (psig):
 - a. Bolts: In accordance with ASTM A307, Grade B.
 - b. Nuts: In accordance with ASTM A563, Grade A.
 - c. Bolts and Nuts: Hot-dip galvanized in accordance with ASTM F2329.
2. On exposed pipes with pressures greater than 150 psig:
 - a. Bolts: In accordance with ASTM A193, Grade B7.

WRAP-IT LINK™

WALL PENETRATION SEALS & WALL SLEEVES WITH WATER STOPS



CCI PIPING SYSTEMS
PIPE PROTECTION PRODUCTS
ccipipe.com • 800.867.2772

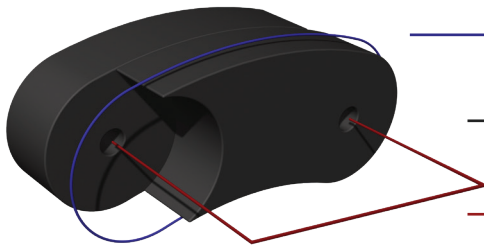
WRAP-IT LINK™

Wrap-It Link™ form a mechanical rubber seal between pipes going through walls, floors, vaults, tanks and pipeline crossings. Wrap-It Link™ make a watertight seal. It can also seal the gap between an inner pipe and an outer pipe sleeve or pipeline casing. It seals the gap between electrical conduits and either the outer conduits or the wall hole it passes through. Wrap-It Link™ are designed to make a hydrostatic seal of up to 20 psig and up to 40 feet of head.



The Wrap-It Link™, in addition to its sealing properties, helps absorb vibration, shock and sound waves. The links will insulate the inner pipe from all other outer structures, including outer pipe sleeves, pipeline crossings, walls, and tanks. It is also ideal for filling gaps in interior walls and floors and keeping unwanted noise and fumes from invading office space. It has the added benefit of diminishing annoying liquid and turbulence noise.

This versatile link system can be utilized to seal the annular space of the carrier pipe and casing pipe applications, as well as wall penetrations of piping system. Wrap-It Link™ have excellent electrical insulation properties and may be used for pressure applications.



• Expanded State Thickness

• Free State Thickness

• Chord Length

WRAP-IT LINK™ CALCULATOR

For your convenience, visit our website to utilize our Wrap-It Link™ calculator. Simply input the outer diameter of the pipe and the inner diameter of the hole. This feature will calculate which model number and how many links your job requires.

PIPE O.D.

HOLE I.D.

☐ INCH

☐ METRIC

PHYSICAL PROPERTIES

Type	Seal Material	Pressure Plates	Bolts & Nuts	Temperature Range (°F)	Features
WL	EPDM Black	Glass Reinforced Plastic	Steel Zinc Dichromate	-40° to +250°	Suitable in most applications in water, both above ground & direct burial. Provides electrical insulation where cathodic protection is required.
WL-SS	EPDM Black	Glass Reinforced Plastic	Stainless Steel (316)	-40° to +250°	Same as WL, but with corrosion resistant stainless steel hardware.
WL-O	Nitrile	Glass Reinforced Plastic	Steel Zinc Dichromate	-40° to +250°	Resistant to most hydrocarbons, oil, gas, jet fuel and many solvents.
WL-O-SS	Nitrile	Glass Reinforced Plastic	Stainless Steel (316)	-40° to +250°	Same as WL-O, but with corrosion resistant stainless steel hardware.

STANDARD WEIGHT STEEL PIPE, PVC AND CPVC PIPE

Nominal Pipe Size	Actual Pipe O.D.	Standard Weight Steel Pipe or PVC Pipe Sleeve*				Cast or Core Bit Drilled Hole*		
		Sleeve Nominal Pipe Size	Sleeve Actual I.D.	Wrap-It Link™ Model No.	Number of Links Required	Hole I.D.	Wrap-It Link™ Model No.	Number of Links Required
1/2"	0.840"	2"	2.067"	WL-200	4	2.0"	WL-200	4
3/4"	1.050"	2-1/2"	2.469"	WL-275	5	2.5"	WL-275	5
1"	1.315"	2-1/2"	2.469"	WL-200	5	3.0"	WL-300	4
1-1/4"	1.660"	3"	3.068"	WL-275	7	3.0"	WL-275	7
1-1/2"	1.900"	3-1/2"	3.548"	WL-300	5	3.5"	WL-300	5
2"	2.375"	4"	4.026"	WL-300	6	4.0"	WL-300	6
2-1/2"	2.875"	4"	4.026"	WL-200	9	4.0"	WL-200	9
3"	3.500"	5"	5.047"	WL-300	8	5.0"	WL-300	8
3-1/2"	4.000"	6"	6.065"	WL-325	5	6.0"	WL-325	5
4"	4.500"	6"	6.065"	WL-300	10	6.0"	WL-300	10
5"	5.563"	8"	7.981"	WL-430	13	8.0"	WL-430	13
6"	6.625"	10"	10.02"	WL-475	10	10.0"	WL-475	10
8"	8.625"	12"	12.00"	WL-475	12	12.0"	WL-475	12
10"	10.750"	14"	13.25"	WL-425	10	14.0"	WL-400	10
12"	12.750"	16"	15.25"	WL-425	12	16.0"	WL-400	12
14"	14.000"	18"	17.25"	WL-400	13	16.0"	WL-325	15
16"	16.000"	20"	19.25"	WL-400	15	18.0"	WL-325	17
18"	18.000"	24"	23.25"	WL-500	16	22.0"	WL-575	20
20"	20.000"	24"	23.25"	WL-400	18	24.0"	WL-575	22
22"	22.000"	26"	25.25"	WL-400	20	26.0"	WL-575	24
24"	24.000"	30"	29.25"	WL-500	21	28.0"	WL-575	26
26"	26.000"	30"	29.25"	WL-400	23	30.0"	WL-575	28
28"	28.000"	34"	33.25"	WL-500	24	32.0"	WL-575	30
30"	30.000"	36"	35.25"	WL-500	26	34.0"	WL-575	32
32"	32.000"	38"	37.25"	WL-500	27	36.0"	WL-575	34
34"	34.000"	40"	39.25"	WL-500	29	38.0"	WL-575	36
36"	36.000"	42"	41.25"	WL-500	30	40.0"	WL-575	38
42"	42.000"	48"	47.25"	WL-500	36	46.0"	WL-575	44
48"	48.000"	54"	53.25"	WL-500	41	52.0"	WL-575	50

*Minimum recommended sleeve length or wall thickness is 4" for Wrap-It Link™ model 325 and smaller and 6"for models 400 and larger.

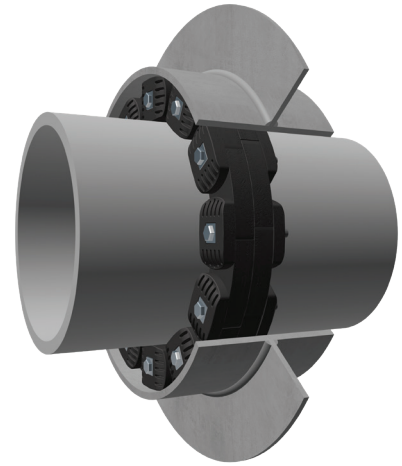
COPPER TUBING

Nominal Pipe Size	Actual Pipe O.D.	Standard Weight Steel Pipe or PVC Pipe Sleeve*				Cast or Core Bit Drilled Hole*		
		Sleeve Nominal Pipe Size	Sleeve Actual I.D.	Wrap-It Link™ Model No.	Number of Links Required	Hole I.D.	Wrap-It Link™ Model No.	Number of Links Required
1/2"	.625"	2"	2.067"	WL-275	4	2.0"	WL-275	4
3/4"	.875"	2"	2.067"	WL-200	4	2.0"	WL-200	4
1"	1.125"	2-1/2"	2.469"	WL-275	6	3.0"	WL-315	4
1-1/4"	1.375"	2-1/2"	2.469"	WL-200	5	3.0"	WL-300	4
1-1/2"	1.625"	3"	3.068"	WL-275	8	3.0"	WL-275	8
2"	2.125"	3-1/2"	3.548"	WL-300	5	3.5"	WL-275	9
2-1/2"	2.625"	4"	4.026"	WL-275	11	4.0"	WL-275	11
3"	3.125"	5"	5.047"	WL-315	8	5.0"	WL-315	8
4"	4.125"	6"	6.065"	WL-315	10	6.0"	WL-315	10
6"	6.125"	8"	7.981"	WL-315	15	8.0"	WL-315	15
8"	8.125"	10"	10.02"	WL-325	9	12.0"	WL-575	10
10"	10.125"	14"	13.25"	WL-410	14	14.0"	WL-575	12
12"	12.125"	16"	15.25"	WL-410	16	16.0"	WL-575	14

*Minimum recommended sleeve length or wall thickness is 4" for Wrap-It Link™ model 325 and smaller and 6"for models 400 and larger.

WRAP-IT LINK™ APPLICATIONS

- Insulated Pipe Seals
- Electrical & Mechanical Contractors
- Offshore Oil Rigs
- Refrigeration Buildings
- Cased Pipeline Crossings
- Precast Concrete Manufacturers
- Valve Pits
- Electrical Isolation of Pipes
- Precast Concrete Vault Seals
- Dual Containment Seals
- Tunneling Operations
- Through Deck Fire Breaks
- Therman Storage Systems
- Pool Contractors
- Wastewater & Water Treatment Plants
- Fire Protection Wall Penetrations
- Telecommunications
- Marine Applications
- Power Generation Dams
- Manhole Pipe Entry Seals
- Perimeter Berm Installations Around Tank Farms
- Noise Dampening
- Bridge Construction



DUCTILE IRON PIPE (AWWA-TYPE)

Nominal Pipe Size	Actual Pipe O.D.	Standard Weight Steel Pipe or PVC Pipe Sleeve*				Cast or Core Bit Drilled Hole*		
		Sleeve Nominal Pipe Size	Sleeve Actual I.D.	Wrap-It Link™ Model No.	Number of Links Required	Hole I.D.	Wrap-It Link™ Model No.	Number of Links Required
2"	2.500"	4"	4.026"	WL-300	6	4.0"	WL-300	6
2-1/4"	2.750"	5"	5.047"	WL-340	7	5.0"	WL-340	7
3"	3.960"	6"	6.065"	WL-325	5	6.0"	WL-325	5
4"	4.800"	8"	7.981"	WL-410	7	8.0"	WL-410	7
6"	6.900"	10"	10.02"	WL-410	10	10.0"	WL-410	10
8"	9.050"	12"	12.00"	WL-400	9	12.0"	WL-400	9
10"	11.100"	14"	13.25"	WL-325	12	14.0"	WL-400	10
12"	13.200"	16"	15.25"	WL-325	14	16.0"	WL-360	21
14"	15.300"	18"	17.25"	WL-325	16	18.0"	WL-425	14
16"	17.400"	20"	19.25"	WL-315	39	20.0"	WL-360	27
18"	19.500"	24"	23.25"	WL-475	25	22.0"	WL-425	18
20"	21.600"	26"	25.25"	WL-475	28	26.0"	WL-525	19
24"	25.800"	30"	29.25"	WL-400	23	30.0"	WL-575	28
30"	32.000"	38"	37.25"	WL-500	27	36.0"	WL-575	34
36"	38.300"	42"	41.25"	WL-400	34	43.0"	WL-525	33
42"	44.500"	48"	47.25"	WL-425	39	49.0"	WL-525	38
48"	50.800"	54"	53.25"	WL-425	45	56.0"	WL-500	43

*Minimum recommended sleeve length or wall thickness is 4" for Wrap-It Link™ model 325 and smaller and 6" for models 400 and larger.

CAST IRON SOIL PIPE (EXTRA HEAVY)

Nominal Pipe Size	Actual Pipe O.D.	Standard Weight Steel Pipe or PVC Pipe Sleeve*				Cast or Core Bit Drilled Hole*		
		Sleeve Nominal Pipe Size	Sleeve Actual I.D.	Wrap-It Link™ Model No.	Number of Links Required	Hole I.D.	Wrap-It Link™ Model No.	Number of Links Required
2"	2.380"	4"	4.026"	WL-300	6	4.0"	WL-300	6
3"	3.500"	5"	5.047"	WL-300	8	5.0"	WL-300	8
4"	4.500"	6"	6.065"	WL-300	10	6.0"	WL-300	10
5"	5.500"	8"	7.981"	WL-425	6	8.0"	WL-425	6
6"	6.500"	10"	10.02"	WL-475	10	10.0"	WL-475	10
8"	8.620"	12"	12.00"	WL-475	12	12.0"	WL-475	12
10"	10.750"	14"	13.25"	WL-425	10	14.0"	WL-400	10
12"	12.750"	16"	15.25"	WL-425	12	16.0"	WL-400	12
15"	15.880"	20"	19.25"	WL-400	15	20.0"	WL-575	18

*Minimum recommended sleeve length or wall thickness is 4" for Wrap-It Link™ model 325 and smaller and 6" for models 400 and larger.

CASING SPACERS & END SEALS

If you are looking to improve performance, reliability and productivity, then CCI's casing spacers are just what you are looking for. Casing spacers provide electrical insulation, will not rot or settle, support carrier pipe weight uniformly, protect against long term corrosion, require no casing filler, and slide into place easily.

Carbon Steel & Stainless Steel Spacers are manufactured using heavy duty two-piece #304 Stainless Steel or Carbon Steel with durable glass filled polymer runners. Spacers are lined with ribbed PVC rubber to prevent electrical contact between the carrier pipe and the metallic spacer. Carbon Steel Spacers are fabricated from pickled and oiled steel and then coated with a fusion bonded copolymer based thermoplastic for superior abrasion resistance and corrosion protection. Spacers for bell & spigot joints, or large diameter applications, are designed and fabricated to insure an extra margin of support. CCI Piping Systems has the experience to design and manufacture virtually any configuration of casing spacer, including multi-carrier clusters within a single casing.

Polyethylene Spacers are injection molded from high-density polyethylene. Providing an economical alternative to metallic casing spacers, they also have excellent dielectric strength and low moisture absorption, which provides minimal electric current loss with no impairment of cathodic protection.

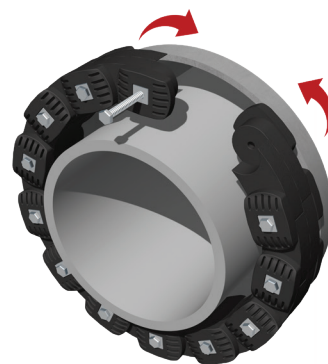
WRAP-IT LINK™ INSTALLATION INSTRUCTIONS

STEP 1

Center the carrier pipe within the core drilled hole, casing pipe, or wall sleeve. The carrier pipe must be centered and adequately supported prior to installing the Wrap-It Link™ belt assembly.

STEP 2

Hand tighten each pressure plate to make slight contact with the link. Wrap the link belt around the carrier pipe and connect the first and last links. Ensure all bolt heads are facing the installer.

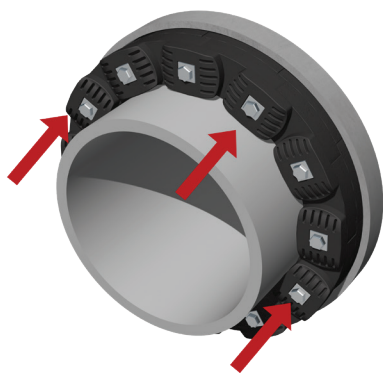


STEP 3

Slide the Wrap-It Link™ belt assembly into the annular space starting at the 6 o'clock position, working both sides up toward the 12 o'clock position. *Note: Slack or sag is normal and the installer should NOT remove any links.*

STEP 4

Using a HAND ratchet with a socket, begin tightening the bolts from the 12 o'clock position and continue clockwise in a sequential pattern.

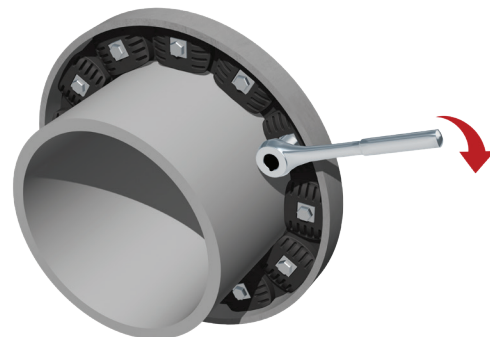


STEP 5

Tighten each link the same number of turns, not to exceed 4 turns per round. Repeat the process until the links begin to bulge slightly around the pressure plates. Make an additional round being careful not to overtighten the bolts.

STEP 6

If a seal is unable to be achieved using the instructions provided, call CCI at 800-867-2772.



CERTIFICATION

CUSTOM PIPING SYSTEMS L.L.C.

PROVIDING CUSTOM FABRICATIONS FOR THE WATER & WASTEWATER INDUSTRIES

9615 Ball Street San Antonio, Texas 78217

Ph: 210-739-0595

Mike@custompipingsystems.com

09/26/2025

C&B Piping (W), Inc.

Attn: Ryan Miller

American Iron and Steel Certification

Contractor: Nan Inc.

Project: HILO HI WWTP

Location: HILO, HAWAII

PO: TBD

I, Mike Kramer, certify that the (melting, bending, coating, galvanizing, cutting, etc.) process for (manufacturing or fabricating) the following products and/or materials shipped or provided for the subject project will be in full compliance with the American Iron and Steel requirement as mandated in EPA's State Revolving Fund Programs.

Item, Products and/or Materials:

- ALL CARBON STEELS HDG PE X PE W/ NO COLLAR WALL SLEEVES PER DET. P302
- ALL 316SS WALL SLEEVES PER DET. P304

Such processes will take place at the following location:

Custom Piping Systems L.L.C., 9615 Ball Street, San Antonio, TX 78217, USA

If any of the above compliance statements change while providing material for this project, we will immediately notify the prime contractor and the engineer.

Mike Kramer

Mike Kramer

QA/QC Manager